LSRII (Sacramento) Standard Operating Procedure & User Guide - STARTUP
UC Davis Flow Cytometry Shared Resource

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Updated: 03/17/2026



1. Power on the LSRII: Note: The lasers on the LSRII require 15-20min of warm up time.
2. While the system is warming up check all liquid levels:
    a. Check that the sheath is filled to the weld line on the tank
        i. To open the sheath tank:
            1. Disconnect the solid green air line from the tank.
            2. Open the vent valve- rotate counterclockwise. **Open just enough to allow the air to release, do NOT open the valve all the way or it can get stuck open.*
            3. Unscrew the tank main nut to the first stage – the nut should spin freely.
            4. The lid should drop down if it does not give it a slight knock.
                a. Be careful not to dislodge the large O-ring around the tank lid.
            5. Remove the lid and place it onto the empty blood bank saline card board box under the LSRII.
        ii. Peek inside the tank once the lid has been removed and check that the liquid is at the top weld line (see red tape on outside of tank for reference).
            1. Refill sheath to weld line if necessary.
        iii. Replace the lid and tighten the main nut.
        **iv. Close the relief valve – turn clockwise until it JUST stops. Do NOT overtighten.**
        v. Re-connect the solid green air pressure line to the tank.
        vi. Open the relief valve once more and listen for the air hiss to let you know the lid is installed correctly.
        vii. **Close the relief valve – turn clockwise until it JUST stops. Do NOT overtighten.**
    b. Check that the waste is less than 50% full,
        i. Life tank or open lid to check levels.
        ii. If the waste is >50%, empty waste tank:
            1. Disconnect the waste tank probe from the waste line @ the quick disconnect. (do NOT remove the probe)
            2. Remove the tank and bring it to the sink.
            3. Place paper town down on counter
            4. Remove the large waste cap and place it into the paper towel.
            5. Carefully dump waste against the back corner of the sink.
            6. **Add 1 liter of 100% bleach to the waste tank** (Use the first plastic line on the outside of the tank as a guide or use the 1 L graduated cylinder above the sink ).
    c. If you had to fill the sheath in 2a or empty the waste in 2b make a note to email Jonathan Van Dyke (jvandyke@health.ucdavis.edu) and inform him as this steps should have been completed by the previous user.

3. Purge air bubbles from system
    a. Obtain the collection beaker and if indeed, empty the collection beaker into the sink.
    b. *Purge air from sheath filter*
        i. Lightly tap the filter on the empty sheath cube at a 45 degree angle in order to move all air bubbles to the filter evacuation tube/port. Open the relief valve and allow sheath to flow into the beaker for 5 seconds or until all air is purged from the filter then close the relief valve.
        ii. If no sheath is flowing from the filter, stop and do not proceed. Check that the tank is properly sealed and try again. Sheath MUST flow from the filter before moving to the next step.
    c. *Purge air from sheath tubing*
        i. Move to the right side of the instrument and open the sheath tubing relief valve and allow sheath to flow into the beaker for 10 seconds, close the relief valve.
        ii. If no sheath is flowing from the relief valve, stop and do not proceed. Check that the tank is properly sealed and try again. Sheath MUST flow from the relief valve before moving to the next step.
    d. *Purge air from inside system*
        i. Remove the water tube from the instrument SIP (leave the tube holder to the side!)
        ii. Press the “Prime” button and wait for the light to return to the standby button, this is one prime cycle. Do this a total of 3 times.
    e. *Purge air from waste tubing*
        i. Replace the tube of DI water on to the SIP.
        ii. Move the tube holder back to the RUN position (closed/under tube).
        iii. Confirm the instrument is set to ‘Low” and press the “Run” button.
        iv. Watch the waste line and ensure all air is pushed into the waste tank and the entire waste line is a solid column of liquid. *note: sometimes this will require flicking the waste line as the bubbles can get caught.
        v. Press the instrument [STBY] button to put the system into StandBy.
4. Empty the collection beaker into the sink.
5. Launch Diva and Log into the software
    a. Log in: *Lab name* P/W: *PI last name*
6. Diva will now try to connect to the LSRII, this could take upwards of 5min. Note: -connecting to cytometer- should be visible on the bottom of the “cytometer window” and in the bottom right of the entire Diva workspace window.
    a. *note: on the rare occasion Diva and the instrument do not connect a complete restart of both the instrument and the computer are required.
7. Once connected a dialog box will often popup stating a mismatch between current settings and CST always select “use CST settings”
    a. *note: On occasion this box will NOT pop up on first log in, this is not a problem as the message will appear after running CST.
8. QC the instrument
    a. Select **Cytometer** –> **CST** from the top menu.

b. Allow CST to connect to the instrument. *note: Diva is now locked to user input as the instrument is now connected with CST
c. Remove air from the droplet containment system.
    i. Ensure a tube of DI water continuing at least 2mL is on the LSRII SIT
    ii. Move the SIT arm to the side and listen for the drop containment pump.
    iii. Allow 1mL of water to be drawn into the instrument.
    iv. Return the SIT arm to the close position.
d. **Go to the refrigerator next to the Fortessa**
    i. Retrieve the blue cardboard CST box labeled “LSRII”
e. If a prepped CST tube is already within the box and has at least 300uL continue, otherwise make a new CST sample.
    i. New CST sample preparation
        1. Vortex the CST bead stock in all 4 orientations (bottom, sideways, upside down, sideways) to fully mix the stock solution.
        2. Place 500uL of DI water into a FACS tube (Just over the weld line at the bottom of the tube)
        3. Place one drop of CST bead stock into the tube
        4. Vortex the CST bead sample.
f. Load the CST sample onto the LSRII SIT.
g. Obtain the current CST bead lot from the CST beads stock tube (do NOT use the lot# on the box)
    i. Confirm/Select the proper bead lot from the dropdown – right side of CST window
h. **Go to the physical LSRII control panel** (front of instrument)
    i. Turn [Fine Adjustment] knob counterclockwise until it stops (can take up to 10 full turns) and then turn it 5 full turns clockwise – to place potentiometer into the middle of its range.
    ii. Press the instrument [Run] button.
**i. Return to LSRII computer and the CST software window**
    i. Press “Run” - in the CST software to start QC. *note: CST will take about 5mins
j. Once CST is completed
    i. Remove the CST beads sample from the SIT
    ii. Put a tube of 2mL DI water on the SIT
    iii. Cap the CST sample tube and return to the CST box.
    iv. Return the CST box to the refrigerator
    v. Click “view report” in popup window to bring up CST report. (Review the report every run!!)
        1. Remember that issues with CST are important to look out for… but ultimately the most important factor is population identification. Errors on channels you are not using are much less of a concern then those you are using.
    vi. In the CST report window/table:
        1. Find bright bead robust CV column and check each primary channel (Blue E, Red C, Violet B)
            a. If CV less than 4.5% - pass no attention needed

b. If CV is > 4.5% but < 6.0% make a note and email JVD
c. If CV is >6.0% try a new bead preparation (unless you made them new today), if this does not solve the issue conduct an abridged cleaning procedure – 10% bleach for 5mins, DI water for 5mins – rerun CST. If problem persists find JVD, do not be afraid to still try samples!

2. Check ΔPMTV (Delta PMT Voltage) for ALL detectors.
    a. If ΔPMTV less than 30+/- pass no attention needed
    b. If ΔPMTV is greater than 30+/- but less than 50+/- make a note and email JVD
    c. If ΔPMTV is 50+/- or greater try a new bead preparation (unless you made them new today), if this does not solve the issue conduct an abridged cleaning procedure – 10% bleach for 5mins, DI water for 5mins – rerun CST. If problem persists find JVD, do not be afraid to still try samples!

3. Check laser delay – this is critical on the LSRII!
    a. Laser delay on FSC for the red laser should be +/- 5 the value reported on the laser delay card above the LSRII computer on the wall. (Subject to change due to alignment changes)
    b. This is critical for determining the stability of the fluidic subsystem.

9. Exit CST
10. Move to the ***DIVA use SOP***.